

CE 521: Ecology and Environment Microbiology

Course Code	CE 521
Course Name	Ecology and Environment Microbiology
L-T-P-C	3-0-2-4
Prerequisites	IC230 (completed a similar Course)
Intended for	PG, PhD
Approval	52nd BoA (Nov 2023)

Course Contents

- **Fundamentals of Ecology:** Basic concepts of ecology and ecosystems, level of organization, ecosystem structure, processes, stability, biogeography and life line zone, population, population characteristic and regulation, species interactions, community and keystone species, succession and concept of climax, fundamental of primary and secondary productivity, energy transfer and nutrient cycling, biodiversity and ecological restoration
- **Introduction of Microbiology:** Science of microbiology, microbial cell and properties, impact of microorganism on humans, historical route of discovery of microbiology, Hooke, van Leeuwenhoek, and Cohn concepts of discovery, cell structure and history; prokaryotic, eukaryotic and viruses, DNA arrangement in microbial cell, metabolic diversity of microbes, Types of bacteria; proteobacteria gram positive and gram negative, archaea, cell morphology and size, cytoplasmic membrane; structure and function, cell wall of prokaryotes; gram positive and gram negative, cell walls of archaea, flagella and motility. (8 Lectures)
- **Environment Microbiology:** Microorganism in the environment, bacterial nutrition and growth, specific growth rate and doubling time, type of culture media; batch and continuous culture, enzyme and catalyst, energy-rich compounds and storage, microbial metabolism-glycolysis, TCA, and ETC, fermentation, nitrogen fixation, nitrogenase, methanogenesis, acetogenesis, biodegradation, bioremediation, bioleaching or biomining, biodegradable organism, environmental factors affecting biodegradation, biodegradation of organic pollutants, biofuel, waste water treatment, water borne microbial diseases, major industrial products, flavouring agents and food supplement, vitamins and beverages, organic acids, aeromicrobiology. (12 Lectures)
- **Himalayan Environmental Microbiology:** Geography and uniqueness of Himalayan region, life in extreme environment, biogeochemical cycle of elements, high altitude lake microbiology, cryospheric microbiology, extremophile (halophile, thermophile, psychrophile) adaptation metabolic processes under extreme conditions, application of cryospheric microbes. (8 Lectures)
- **Advanced Microbiology Tools and Techniques:** Light microbiology and its principles, compound light microscope, increasing contrast in light microscopy, phase contrast and microfield microscopy, fluorescence microscopy, differential interference contrast microscopy (DIC), atomic force microscopy (AFM), confocal scanning Laser microscopy (CSLM), Electron microscopy, OMIC techniques in microbial remediation processes, genomics, proteomics, and metabolomics in microbial remediation, recent advances in in silico approaches for the removal of environmental pollution.

Laboratory Modules

1. Method of vegetation sampling, quadrat method; shape, size and number, density, relative density, frequency, relative frequency of vegetation (6 Lectures)
2. Basic instrumentation, safety and disinfection and in microbiology laboratory (2 Lectures)
3. Preparation and sterilization of various growth media for microbial isolation (2 Lectures)
4. Microbial isolation and maintenance through serial dilution, plating, streaking, slant and storage of microbial culture (4 Lectures)
5. Morphological and biochemical characterization: Gram staining, catalase, urease, oxidase, indole, methyl red etc. (6 Lectures)
6. Growth curve, effect of physical parameter such as temperature, pH, carbon and nitrogen sources on microbial growth (4 Lectures)
7. Isolation of environmental and industrially important microbes (2 Lectures)
8. Isolation of halophile, thermophile, psychrophile from Himalayan sources. (2 Lectures)

Text Books

1. Madigan, M. T., Clark, D.P., Stahl, D., & Martinko, J. M., **Brock biology of microorganisms**, 13th Edition, Benjamin Cummings, 2010.
2. Singh, J. S., Singh, S. P., & Gupta, S. R., **Ecology, environmental science & conservation**, S. Chand Publishing, 2014.

References

1. Dhakar, K and Pandey, A., **Microbial ecology from the Himalayan cryosphere perspective**, microorganisms, 8(2) 257, 2020
2. Liu, Y., Yao, T., Jiao, N., Tian, L., Hu, A., Yu, W., and Li, S., **Microbial diversity in the snow, a moraine lake and a stream in Himalayan Glacia**, Extremophiles, 15, 411-421, 2011.
3. Sati, S. C. and Belwal, M., **Microbes: Diversity and Biotechnology**, Daya Publishing House.
4. Maier, R. M., Pepper, I., and Gerba, C. P., **Environmental Microbiology A laboratory Manual**, Academic Press.
5. Odum, E. ., and Barrett, G. W., **Fundamentals of Ecology**, 5th Edition, Cengage Learning India, 1971.
6. Pepper, I. L., Gerba, C. P., Gentry, T. J., and Maier, R. M. (editors), **Environmental microbiology**, Academic Press, 2011.
7. Giri, A., and Pant, D., **CO2 management using carbonic anhydrase producing microbes from western Indian Himalaya**, Bioresource Technology Reports, 3, 100320.